# Wave Motion

Prepared by:

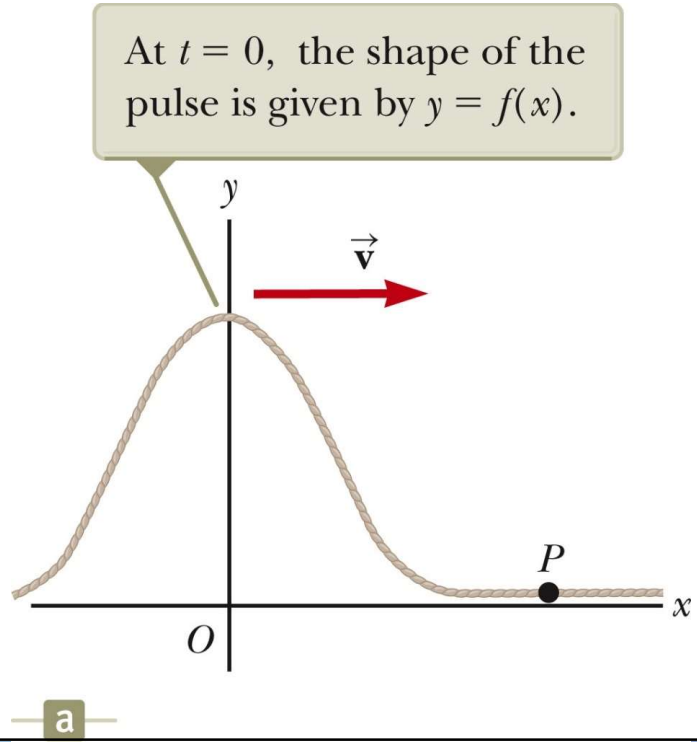
Dr. Yasser El-Batawy

# Wave Motion



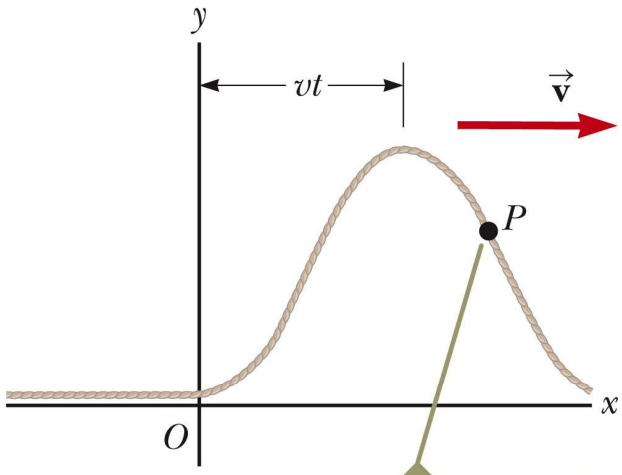
# Traveling Pulse

- The shape of the pulse at $t = 0$ is shown.
- The shape can be represented by some mathematical function, $y\,(x,0) = f\,(x)$.
  - This describes the transverse position y of the element of the string located at each value of x at $t = 0$.



# Traveling Pulse

- The speed of the pulse is $v$.
- At some time, $t$, the pulse has traveled a distance $v\,t$.
- The shape of the pulse does not change with time.
- Its vertical position is now $y = f(x - vt)$.




At some later time $t$, the shape of the pulse remains unchanged and the vertical position of an element of the medium at any point $P$ is given by $y = f(x - vt)$.

# Traveling Pulse

- For a pulse traveling to the right, then $y\,(x,\,t) = f\,(x - vt)$
- For a pulse traveling to the left, then $y\,(x,\,t) = f\,(x + vt)$
- The function y is also called the **wave function**: $y\,(x,\,t)$.
- The wave function represents the y coordinate of any element located at position x at any time *t*.
  - The y coordinate is the transverse position.
- If *t* is fixed then the wave function is called the **waveform.**
  - It defines a curve representing the geometric shape of the pulse at that time.

# The Linear Wave Equation

- The wave functions $y(x, t)$ represent solutions of an equation called the **linear wave equation.**
- This equation gives a complete description of the wave motion.
- From it you can determine the wave speed
- The linear wave equation is basic to many forms of wave motion.

# Linear Wave Equation

- The linear wave equation is satisfied by any wave function having the form

$$y = f(x \pm vt)$$

- The linear wave equation is also a direct consequence of Newton's Second Law applied to any element of a string carrying a traveling wave.

# Wave Equation

$$y = f(x \pm vt)$$

$$\frac{\partial y}{\partial x} = f'(x \pm vt) \quad \Rightarrow \quad \frac{\partial^2 y}{\partial^2 x} = f''(x \pm vt)$$

$$\frac{\partial y}{\partial t} = \pm v f'(x \pm vt) \quad \Rightarrow \quad \frac{\partial^2 y}{\partial^2 t} = (\pm v)^2 f''(x \pm vt) = v^2 f''$$

$$\frac{\partial^2 y}{\partial^2 x} = \frac{1}{v^2}\frac{\partial^2 y}{\partial^2 t}$$

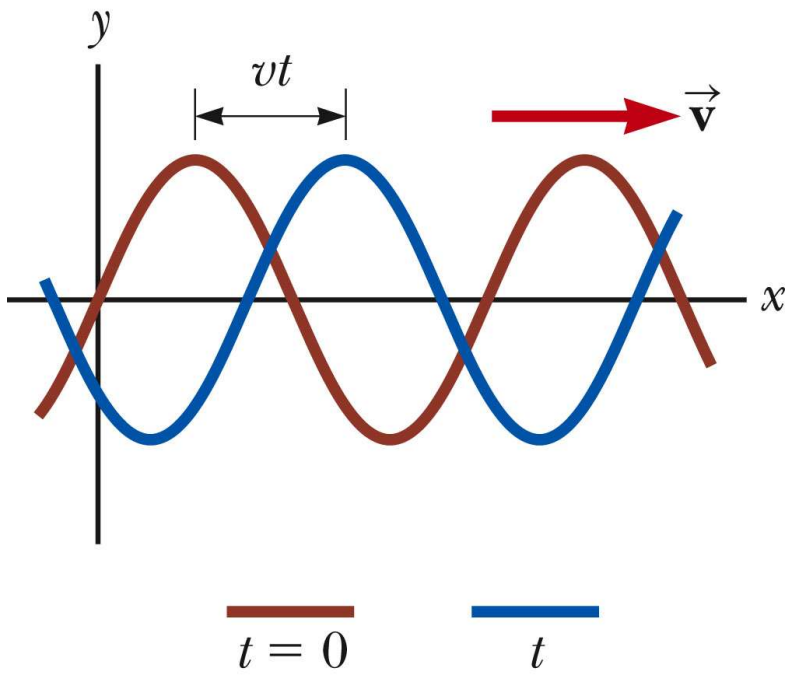
# Linear Wave Equation, General

- The equation can be written as $\frac{\partial^2 y}{\partial^2 x} = \frac{1}{v^2}\frac{\partial^2 y}{\partial^2 t}$
- This applies in general to various types of traveling waves.
  - y represents various positions.
    - For a string, it is the vertical displacement of the elements of the string.
    - For a sound wave propagating through a gas, it is the longitudinal position of the elements of the gas from the equilibrium position.
    - For electromagnetic waves, it is the electric or magnetic field components.

# Traveling Wave

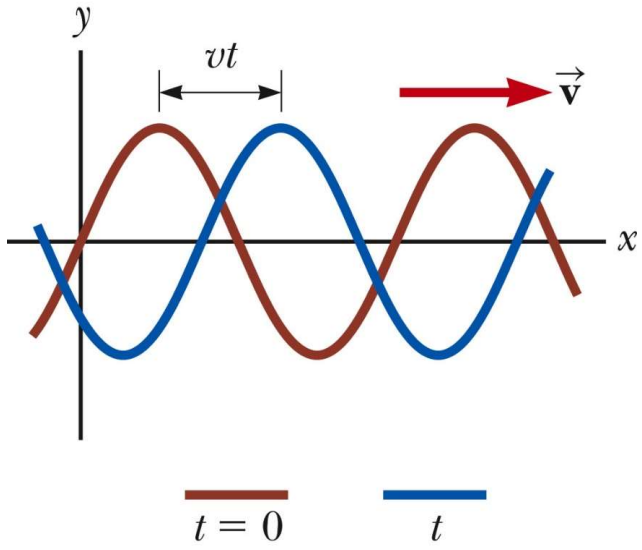
- The wave represented by the curve shown is a **sinusoidal wave.**

- This is the simplest example of a periodic continuous wave.
  - It can be used to build more complex waves.



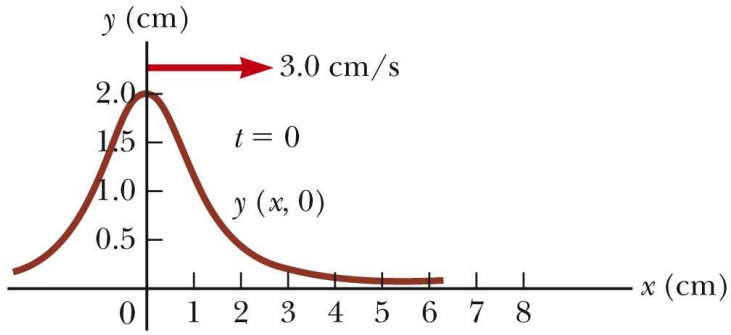
# Traveling Waves

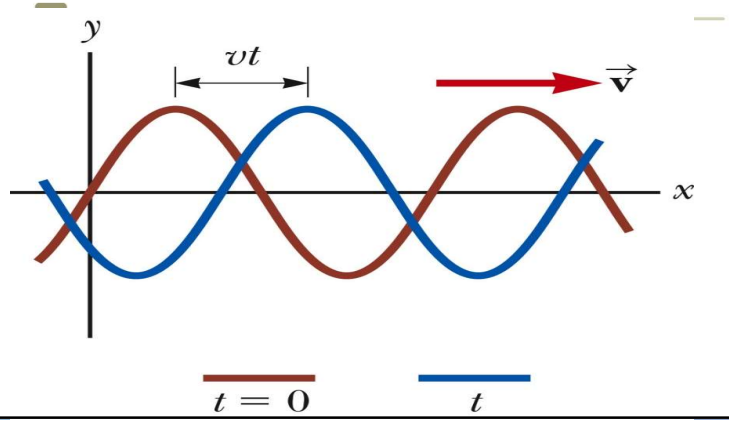
- The wave moves toward the right.
- Each element moves up and down in simple harmonic motion.
  - This is the motion of the elements of the medium.
- It is important to distinguish between the motion of the wave and the motion of the elements of the medium.



# Terminology: Amplitude and Wavelength

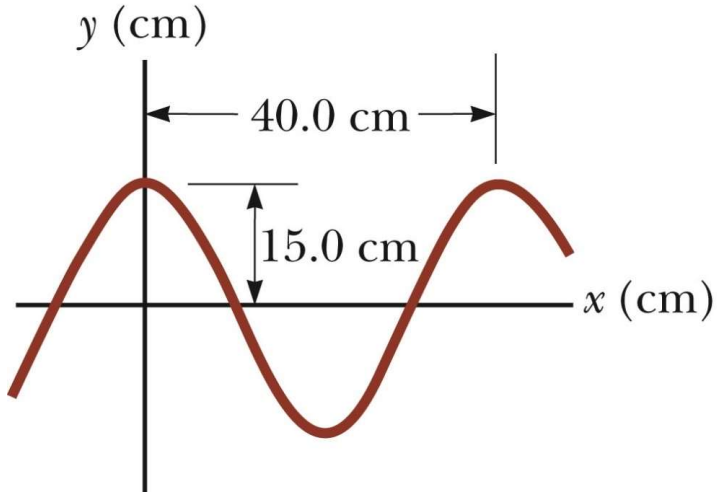
- The **crest** of the wave is the location of the maximum displacement of the element from its normal position.
  - This distance is called the **amplitude**, *A*.
- The **wavelength**, $\lambda$, is the distance from one crest to the next.
  - More generally, the wavelength is the minimum distance between any two identical points on adjacent waves.



# Terminology: Amplitude and Wavelength

- The wavelength, $\lambda$, is 40.0 cm
- The amplitude, $A$, is 15.0 cm



# Terminology: Period and Frequency

- The **period,** T , is the time interval required for two identical points of adjacent waves to pass by a point.
  - The period of the wave is the same as the period of the simple harmonic oscillation of one element of the medium.

- The **frequency**, $f$, is the number of crests (or any point on the wave) that pass a given point in a unit time interval.
  - The time interval is most commonly the second.
  - The frequency of the wave is the same as the frequency of the simple harmonic motion of one element of the medium.

# Terminology: Period and Frequency

- The ***period***, *T*, of the motion is the time interval required for the particle to go through one full cycle of its motion. The inverse of the period is called the ***frequency.***

$$T = \frac{2\pi}{\omega}$$

- The frequency represents the number of oscillations that the particle undergoes per unit time interval.
- Units are cycles per second = hertz (Hz).

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

# Speed of Waves

- Waves travel with a specific speed.
  - The speed depends on the properties of the medium being disturbed.
- The wave function is given by

$$y(x,t) = A\sin\left[K\left(x \pm vt\right) + \alpha\right]$$

  - This is for a wave moving to the right.
  - For a wave moving to the left, replace $x - vt$ with $x + vt$.

# Traveling Wave

- $y(x,t) = A \sin[\, k\,(x-vt) + \alpha\,]$
- A = amplitude of the wave
- At certain $t$, $y(x,t) = y(x + \lambda, t)$ , as the wave repeats itself every $\lambda$
- $A \sin[\, k\,(x-vt) + \alpha\,] = A \sin[\, k\,(\, x + \lambda - vt\,) + \alpha\,]$
- $kx = 2\pi$ , where k is the propagation const. and is given by

$$k = \frac{2\pi}{\lambda}$$

# Traveling Wave

- At certain $x$, $y(x,t) = y(x, t+T)$ , as the particle repeats its motion every $T$
- $A \sin[\, k\,(x-vt) + \alpha\,] = A \sin[\, k\,(\, x - v(t+T)\,) + \alpha\,]$
- $kvT = 2\pi$, then $kv = 2\pi / T = \omega$

$$v = \frac{\omega}{k} = \lambda f$$

- The wave equation can be written as, where $\alpha$ is the phase constant

$$y(x,t) = A \sin[\omega t \pm kx + \alpha\,]$$

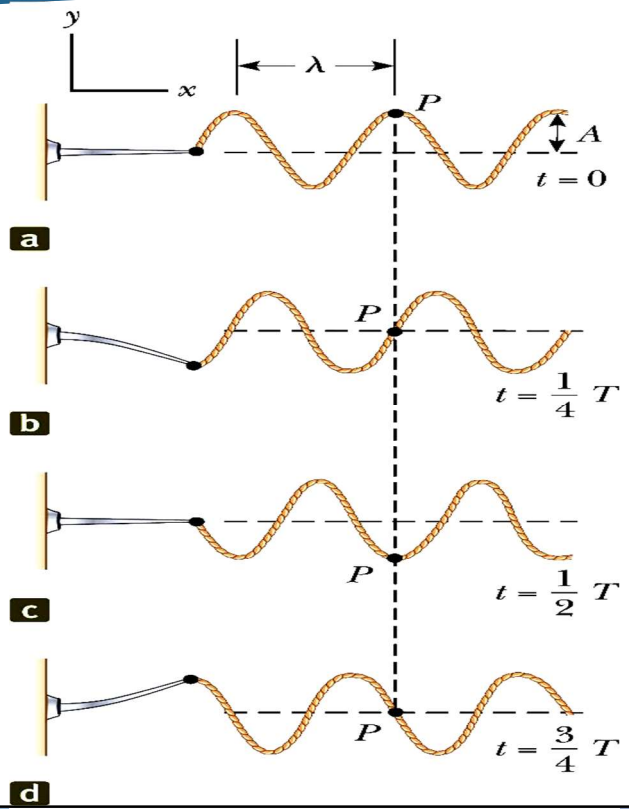
# Transverse Velocity & Acceleration

- The transverse speed of the element is $v_y = \left.\frac{dy}{dt}\right]_{x=\text{constant}}$
- or $v_y = -\omega A \cos(kx - \omega t)$
  - $v_{max} = \omega A$
- This is different than the speed of the wave itself.

- The transverse acceleration of the element is $a_y = \left.\frac{dv_y}{dt}\right]_{x=\text{constant}}$
- or $a_y = -\omega^2 A \sin(kx - \omega t)$
  - $a_{max} = \omega^2 A$

# Sinusoidal Wave on a String

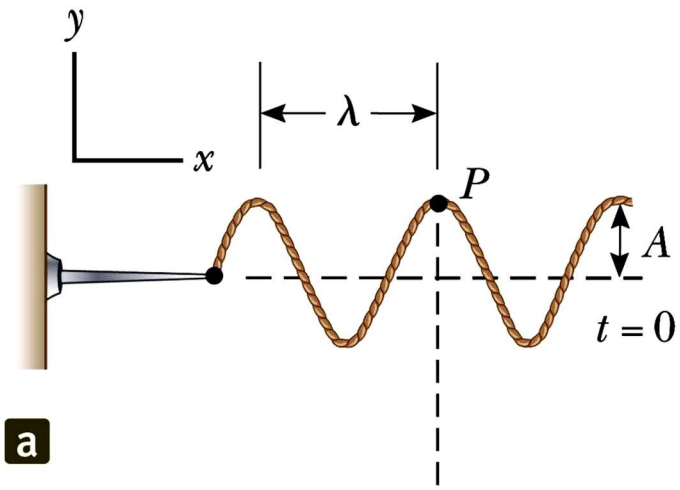
- To create a series of pulses, the string can be attached to an oscillating blade.
- The wave consists of a series of identical waveforms.
- The relationships between speed, velocity, and period hold.



# Sinusoidal Wave on a String

- Each element of the string oscillates vertically with simple harmonic motion. vibrating with a frequency equal to the frequency of the oscillation of the blade.

- At $t = 0$, $y = A \sin(k x - \omega t)$

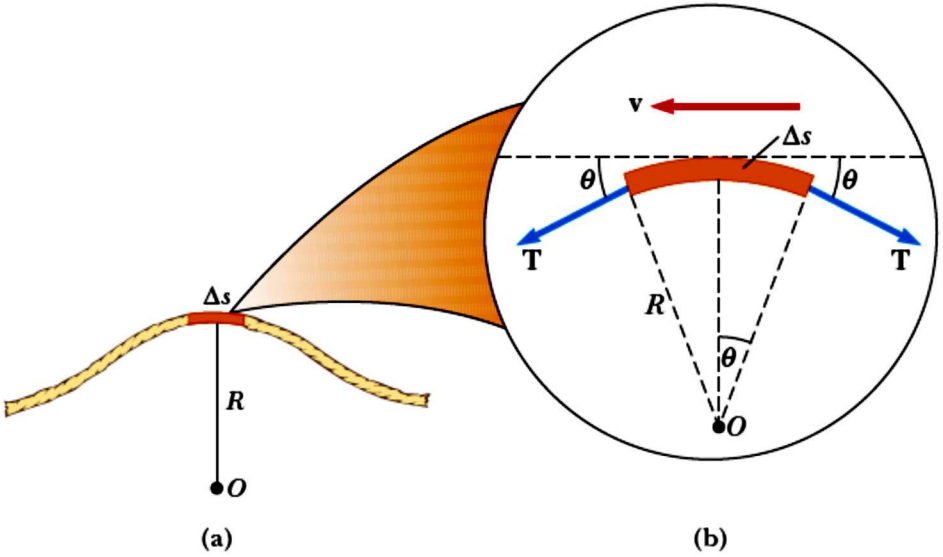


# Wave on a string

- The string is under tension *T*.
- Consider one small string element of length $\Delta x$.
- The net force acting in the *y* direction is .

$$\Sigma F_y \approx 2T \sin\theta$$

  - This uses the small-angle approximation that sin θ ≈ tan θ.



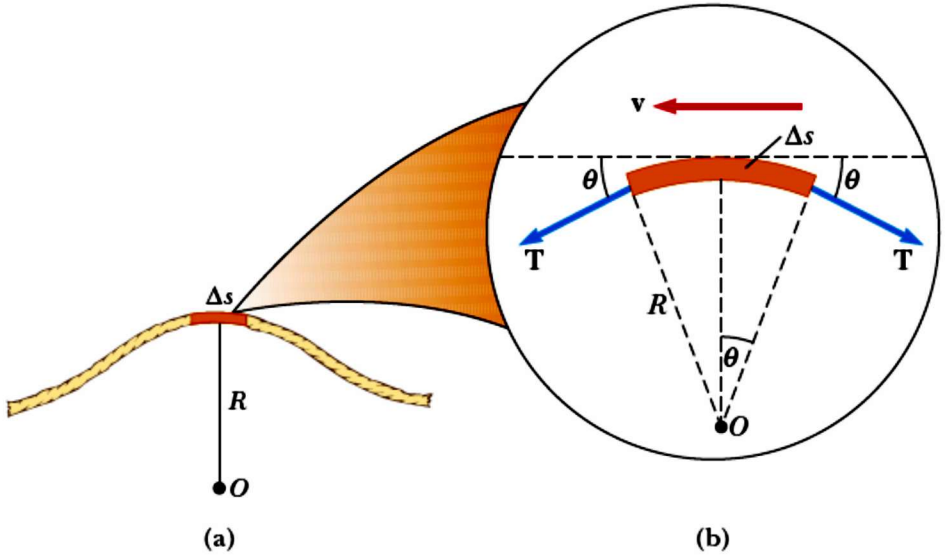
# Wave on a string

$$F_c \approx 2T\theta \qquad m = \mu\Delta s = \mu(2R\theta)$$

$$F_c = ma_c = \mu\Delta s\,\frac{v^2}{R}$$

$$\therefore 2T\,\theta = 2\mu R\,\theta\,\frac{v^2}{R}$$

$$\therefore v = \sqrt{\frac{T}{\mu}}$$



# Speed of a Wave on a String

- The speed of the wave depends on the physical characteristics of the string and the tension to which the string is subjected.

$$v = \sqrt{\frac{\text{tension}}{\text{mass/length}}} = \sqrt{\frac{T}{\mu}}$$

- This assumes that the tension is not affected by the pulse.
- This does not assume any particular shape for the pulse.

# Energy in Waves

- Waves transport energy when they propagate through a medium.
- We can model each element of a string as a simple harmonic oscillator.
- Every element has the same total energy.
- Each element can be considered to have a mass of *dm*.
- The total energy associated with the motion of the element is

$$dE = \tfrac{1}{2}\,(dm)\,\omega^2 A^2.$$

- The mass $dm = \mu dx = \rho\, dvol,$
  - *where* $\mu$ = mass/ unit length, $\rho$ = density of the medium

# Power Associated with a Wave

- The power is the rate at which the energy is being transferred:

$$P = \frac{E_\lambda}{T} = \frac{\frac{1}{2}\mu\omega^2 A^2 \lambda}{T} = \frac{1}{2}\mu\omega^2 A^2 v$$

- The power transfer by a sinusoidal wave on a string is proportional to the
  - Square of the frequency
  - Square of the amplitude
  - Wave speed

The rate of energy transfer in any sinusoidal wave is proportional to the square of the angular frequency and to the square of the amplitude.

# Intensity of waves

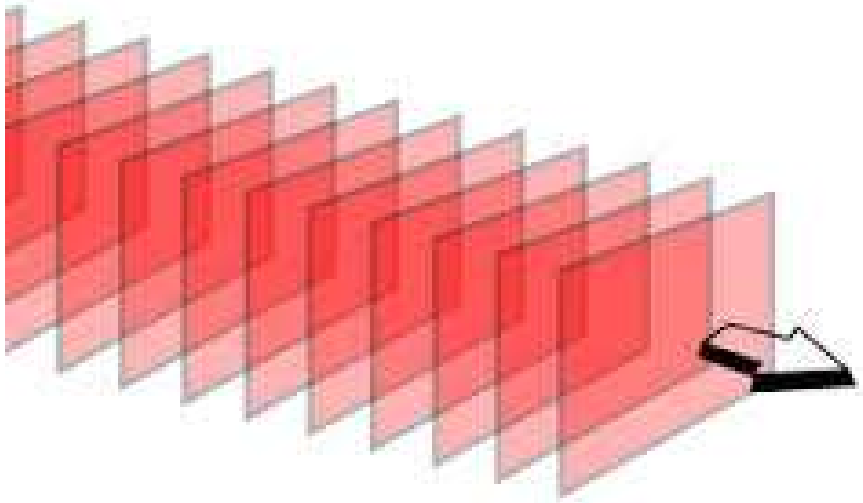
- Intensity of a wave "$I$" is defined as the power associated with unit cross sectional area of the propagation.
- Then

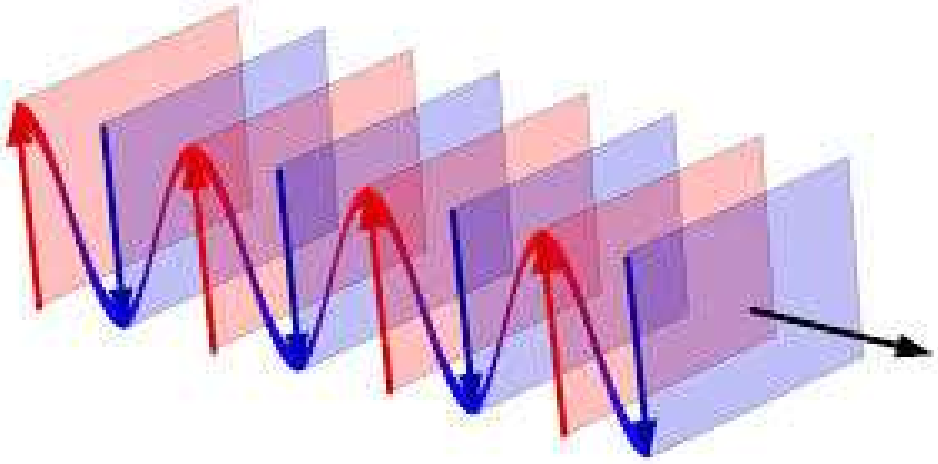
$$I = \frac{dE}{dt * dArea} = \frac{\left(\frac{1}{2}\right) dm \;\; {}^{2} A^{2}}{dt * dArea}$$

$$I = \frac{1}{2} \rho v \omega^{2} A^{2}$$

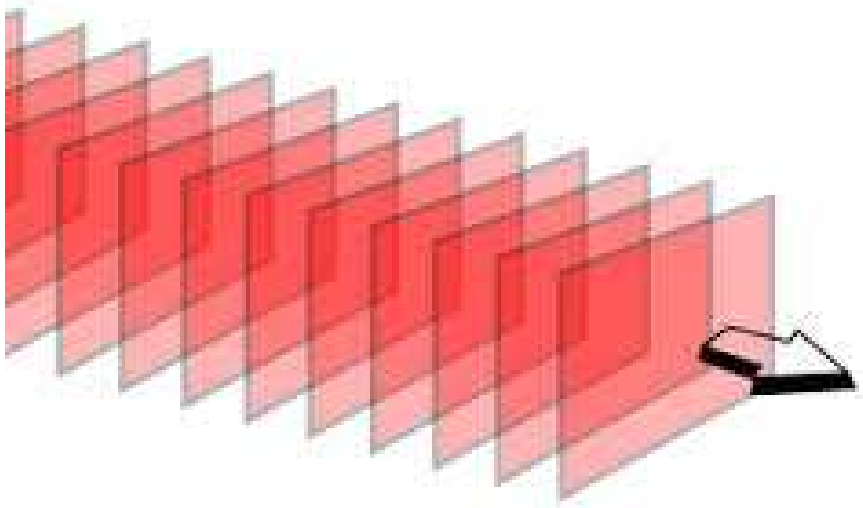
- $\omega$ and $A$ are related to the source of the wave
- where $\rho$ and $v$ are related to the medium of the propagation

# Planar Wave





# Energy in Planar Waves

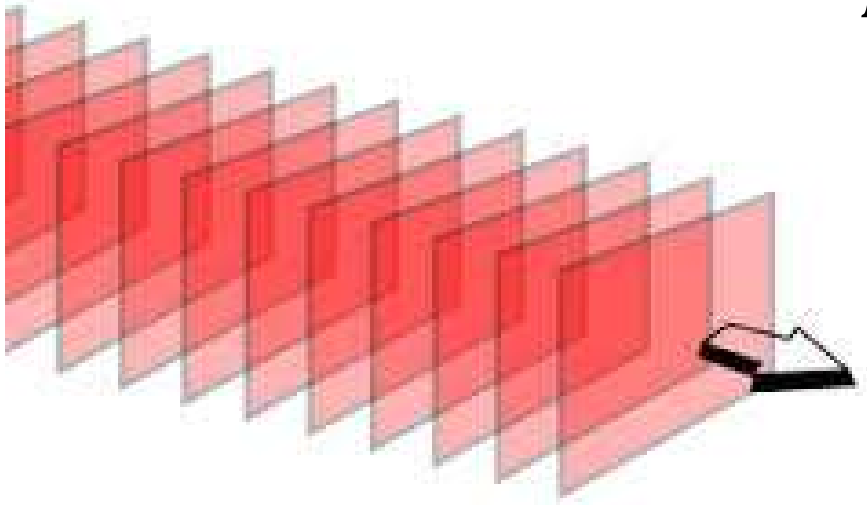


If a wave is able to spread out in one direction perpendicular to the planar source and the medium is uniform, the wave is planar.

Just from geometrical considerations, as long as the power output is constant, we see:

$$\boldsymbol{I} = Constant$$

# Planar Waves

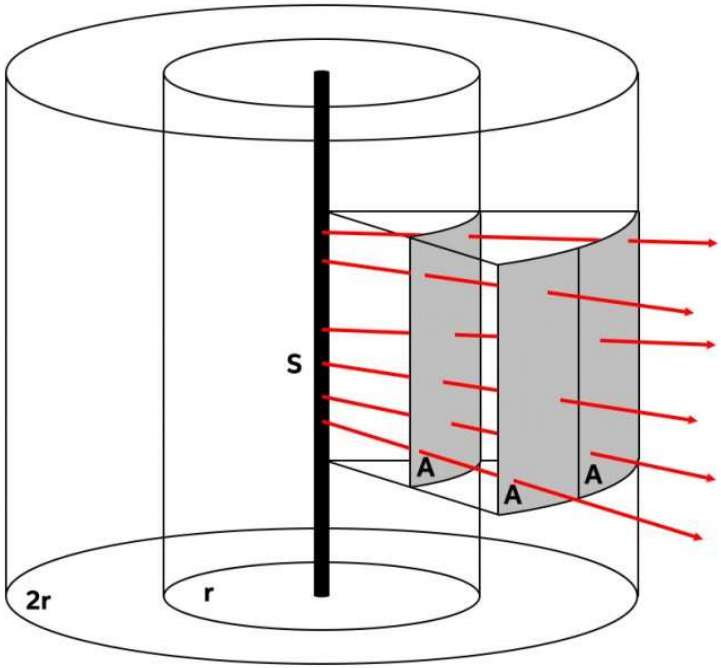


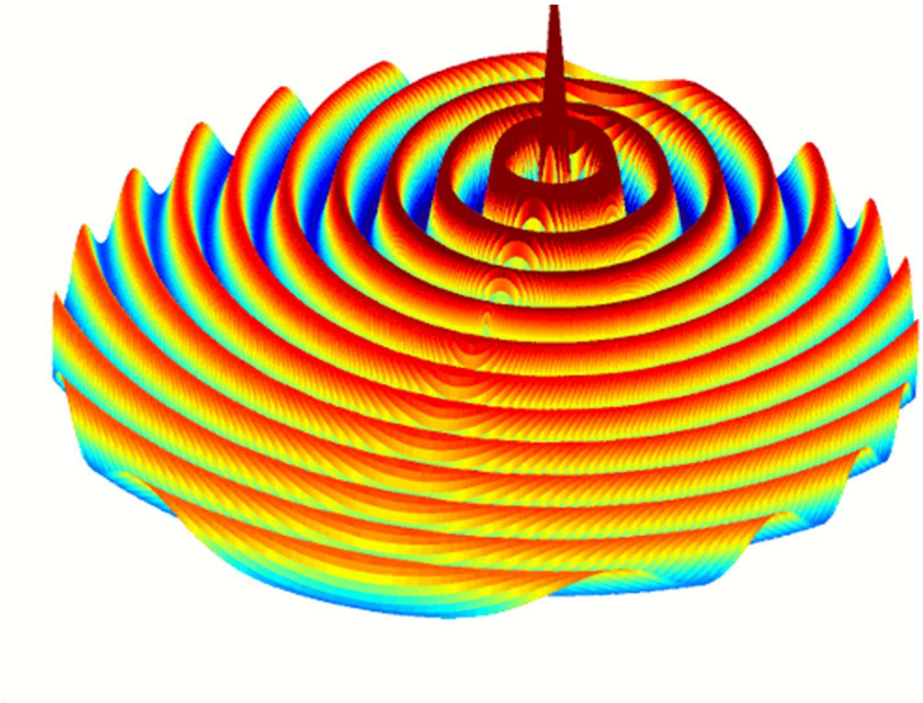
$$I \propto const. \Rightarrow Amplitude = const.$$

- The direction of the wave can be either in +ve or -ve axial directions, then the planar wave equation can be written as

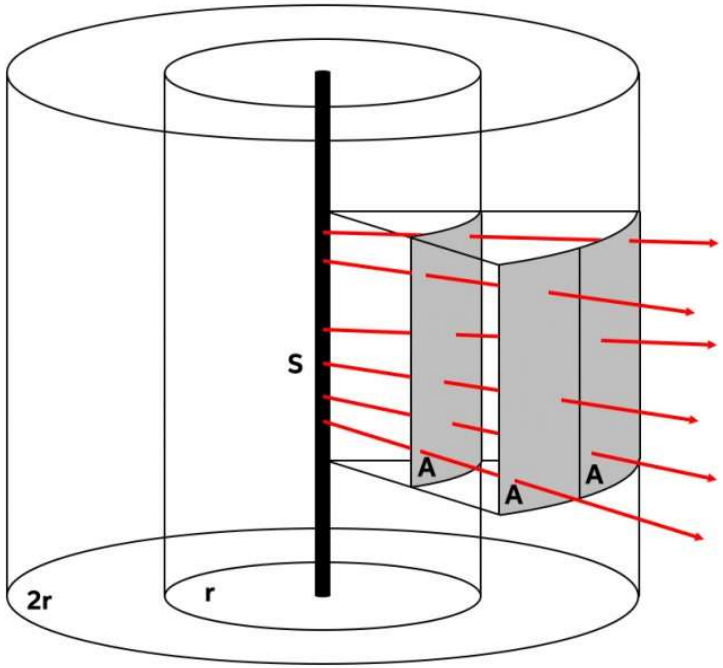
$$\emptyset(x,t) = A\sin(\omega t \pm kx + \alpha)$$

# Cylindrical Wave



# Energy in Cylindrical Waves

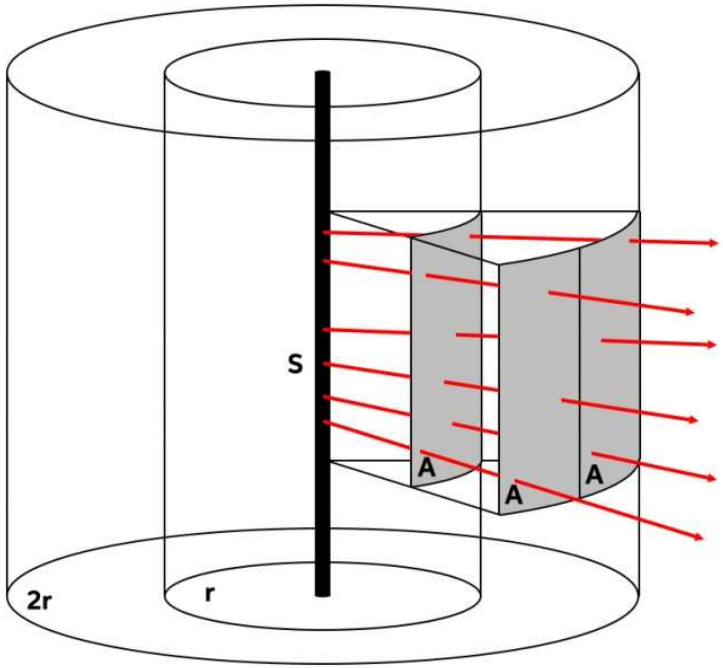



If a wave is able to spread from the line source, and the medium is uniform, the wave is cylindrical.

Just from geometrical considerations, as long as the power output is constant, we see:

$$I \propto \frac{1}{r}$$

# Cylindrical Wave

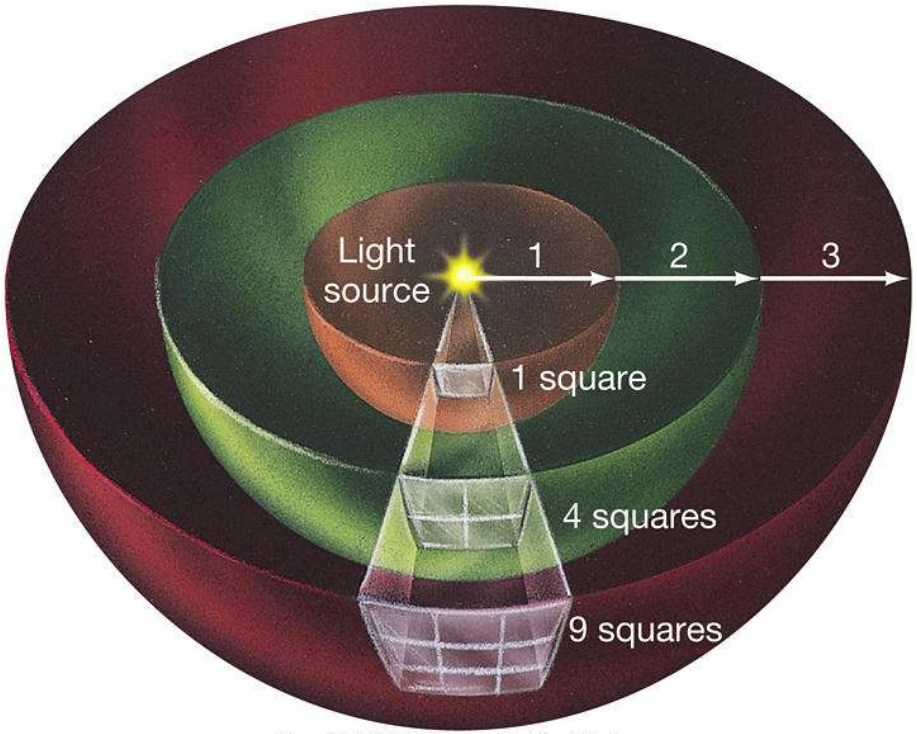



$$I \propto \frac{1}{r} \quad \Rightarrow \quad Amplitude \propto \frac{1}{\sqrt{r}}$$

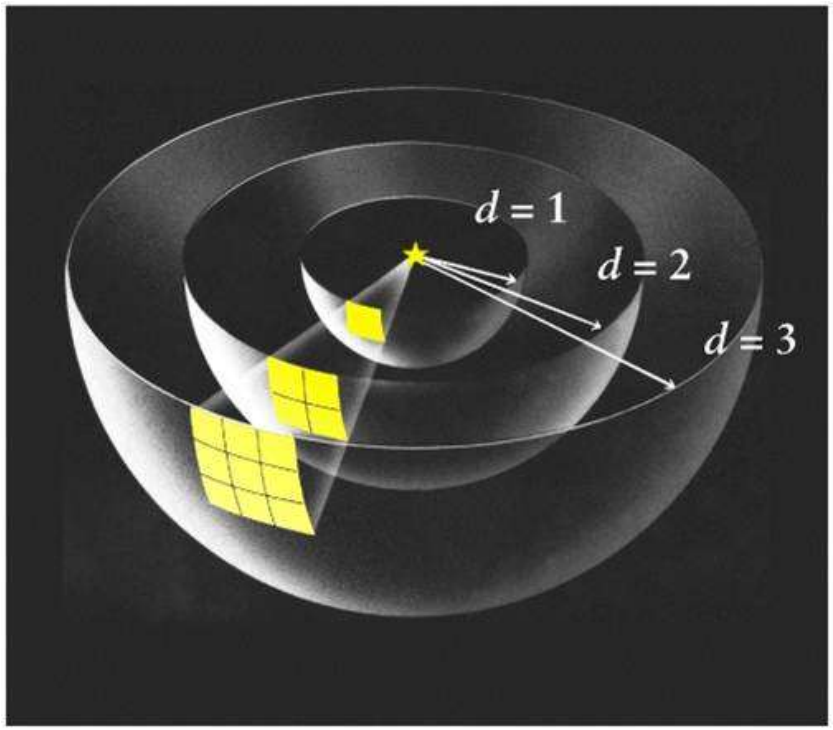
- The direction is only in the +ive radial direction, then the wave equation can be written as

$$\emptyset(r,t) = \frac{A}{\sqrt{r}}\sin(\omega t - kr + \alpha)$$

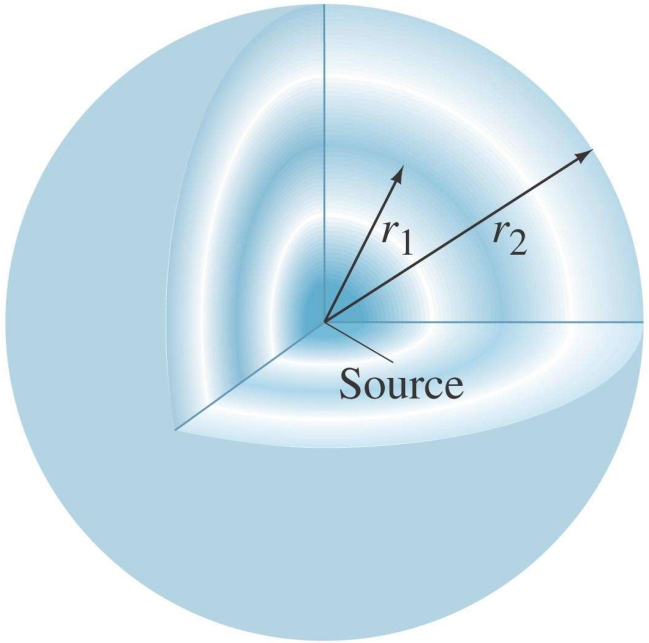
# Spherical Waves




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# Energy in Spherical Waves

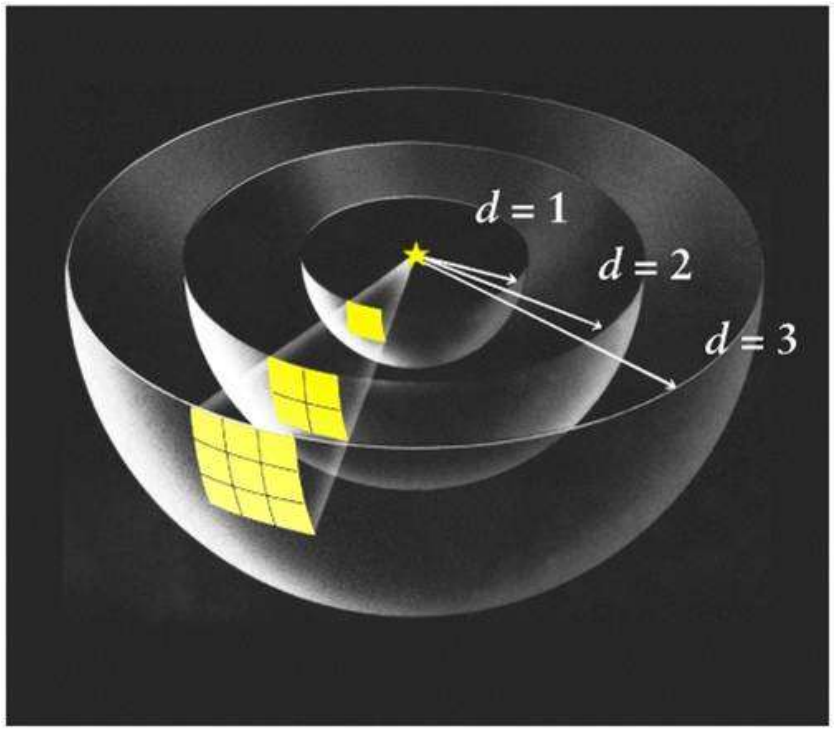



If a wave is able to spread out three-dimensionally from its source, and the medium is uniform, the wave is spherical.

Just from geometrical considerations, as long as the power output is constant, we see:

$$I \propto \frac{1}{r^2}.$$

# Spherical Waves




$$I \propto \frac{1}{r^2} \quad \Rightarrow \quad Amplitude \propto \frac{1}{r}$$

- The direction is only in the +ve radial direction, then the wave equation can be written as

$$\emptyset(r,t) = \frac{A}{r}\sin(\omega t - kr + \alpha)$$

# Superposition of Waves

- If we have many waves with the same wavelength and frequency but differ in Amplitudes and phase constants
- Then, the resultant will be with the same wavelength and frequency.
- How we can get the amplitude and the phase constant of the resultant wave ?

# Superposition of Waves

- $E_1 = E_{o1}\sin(\omega t - kx + \alpha_1)$
- $E_2 = E_{o2}\sin(\omega t - kx + \alpha_2)$
- …
- $E_N = E_{oN}\sin(\omega t - kx + \alpha_N)$
- The resultant wave will be expressed as
- $E_T = E_{oT}\sin(\omega t - kx + \alpha_T)$

# Superposition of Waves

- All these waves can be expressed as
- $E_1 = E_{o1}\sin(\omega t\text{-}kx+\alpha_1) = E_{o1}\cos(\omega t\text{-}kx)\sin(\alpha_1) + E_{o1}\sin(\omega t\text{-}kx)\cos(\alpha_1)$
- $E_2 = E_{o2}\sin(\omega t\text{-}kx+\alpha_2) = E_{o2}\cos(\omega t\text{-}kx)\sin(\alpha_2) + E_{o2}\sin(\omega t\text{-}kx)\cos(\alpha_2)$
- …
- $E_N = E_{o\mathrm{N}}\sin(\omega t\text{-}kx+\alpha_N) = E_{o\mathrm{N}}\cos(\omega t\text{-}kx)\sin(\alpha_\mathrm{N}) + E_{o\mathrm{N}}\sin(\omega t\text{-}kx)\cos(\alpha_\mathrm{N})$

and the resultant wave will be expressed as

- $E_\mathrm{T} = E_{oT}\sin(\omega t\text{-}kx+\alpha_T) = E_{o\mathrm{T}}\cos(\omega t\text{-}kx)\sin(\alpha_\mathrm{T}) + E_{o\mathrm{T}}\sin(\omega t\text{-}kx)\cos(\alpha_T)$

# Superposition of Waves

- Then;
- $E_{0T}\cos(\alpha_T) = \sum_{i=1}^{N} E_{0i}\cos(\alpha_i)$
- $E_{0T}\sin(\alpha_T) = \sum_{i=1}^{N} E_{0i}\sin(\alpha_i)$
- Then; $E_{oT}^2 = \left(\sum_{i=1}^{N} E_{0i}\sin(\alpha_i)\right)^2 + \left(\sum_{i=1}^{N} E_{0i}\cos(\alpha_i)\right)^2$

$$\tan(\alpha_T) = \frac{\sum_{i=1}^{N} E_{0i}\sin(\alpha_i)}{\sum_{i=1}^{N} E_{0i}\cos(\alpha_i)}$$

# Example 1

- A transverse sinusoidal wave on a string has a period $T$ = 25 ms and travels in the negative x direction with a speed of 30 m/s. At $t$ = 0, a particle on the string at x = 0 has a transverse position of 2 cm and is traveling downward with a speed of 2 m/s.
  - What is the amplitude of the wave?
  - What is the initial phase angle?
  - What is the maximum transverse speed of the string?
  - Write the wave function for the wave.

**(Answer: 0.0215m, 1.95 rad, 5.41 m/s)**

# Example 2

- A stone is thrown into a quiet pool of water. With no fluid friction, the amplitude of the waves falls off with distance *r* from the impact point as

**Solution: $1/\sqrt{r}$**

# Example 3

- A boy has climbed to the top of a 5 m tall tree. When he shouts, the sound waves have an intensity of 0.25W/$m^2$ at a 1.00 m distance from the top of the tree. The ratio of intensity at the base of the tree to the intensity 1 m from the top of the tree is ?

**Solution: 1/25**